

Data Preparation Analysis

Project - Proposal & Outline

Topic: Diabetes Risk Prediction using Machine Learning

Group Members

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1. Project Proposal

Formal Description & Research Goal

This project aims to develop a predictive analytics framework for assessing diabetes risk using machine learning. The goal is to identify key health indicators contributing to diabetes and build a model that facilitates early detection and preventive interventions.

Research Questions

1. What are the most significant health indicators influencing diabetes risk?
2. How effective are different machine learning algorithms in predicting diabetes?
3. Can modifiable risk factors (e.g., diet, exercise) improve prediction accuracy?
4. How well do predictive models generalize across diverse populations?
5. How can model interpretability be improved to provide meaningful insights for healthcare professionals?

Proposed Methodology

- Data Preprocessing: Handle missing values, outliers, and data imbalances.
- Feature Selection: Use correlation analysis and mutual information to identify key features.
- Model Development: Train and compare multiple classification models, including Logistic Regression as a baseline for interpretability, Decision Trees and Random Forest for handling non-linearity and feature importance, and Neural Networks for complex pattern recognition.
- Evaluation & Validation: Assess models using accuracy, precision, recall, and AUC-ROC, with cross-validation for reliability.

Metrics for Analysis

- Accuracy: Measures overall correctness.
- Precision & Recall: Evaluates classification performance.
- AUC-ROC: Measures the model's ability to distinguish between diabetic and non-diabetic cases.

2. Project Outline

Literature Review & Related Work

- ["Diabetes Prediction using Machine Learning Algorithms"](#)
 - The study explores the application of machine learning techniques for diabetes prediction, addressing the limitations of traditional diagnostic methods. The paper highlights that diabetes mellitus (DM) is a growing health concern influenced by factors such as age, obesity, lifestyle, and hereditary conditions. The authors review prior research on machine learning-based diabetes prediction, discussing models like Decision Trees, Naïve

Bayes, Artificial Neural Networks, and Random Forest. Their proposed approach involves five key stages: dataset collection, preprocessing, clustering using K-means, model training with various machine learning algorithms, and evaluation using accuracy and F1-score. The study finds that Logistic Regression achieves the highest accuracy (97.2% with pipelining) and outperforms predictions using the PIMA dataset.

- [“A survey on diabetes risk prediction using machine learning approaches”](#)
 - The paper provides an overview of different machine learning methods used to predict the risk of diabetes. It highlights the rising number of diabetes cases and the serious health issues it causes, stressing how important it is to detect the disease early for better management. The study covers several machine learning techniques, including Support Vector Machines (SVM), K-Nearest Neighbours (KNN), Random Forest (RF), Decision Trees, Naïve Bayes, and Artificial Neural Networks. These methods have shown good results in predicting diabetes. The paper also divides these techniques into supervised and unsupervised learning and reviews how well they perform on popular datasets like the PIMA Indian Diabetes dataset. It shows that SVM and Random Forest are particularly effective, consistently achieving high accuracy. The study also emphasizes the need to improve these models so they can work better and more accurately across different groups of people.
- [“A model for early prediction of diabetes”](#)
 - The paper looks at how machine learning can help detect diabetes early. Since diabetes is becoming more common worldwide, catching it early can make a big difference in treatment and health outcomes. The study uses the PIMA Indian Diabetes dataset, which has 768 records with 9 different health-related factors. The researchers test different machine learning models, including Artificial Neural Networks (ANN), Random Forest (RF), and K-means clustering. They also clean the data and use techniques like Principal Component Analysis (PCA) to simplify it before training the models. Among these models, ANN performed the best, with an accuracy of 75.7%, followed by Random Forest at 74.7% and K-means at 73.6%. The study also used the Apriori algorithm to find patterns in the data, showing strong links between diabetes and factors like BMI and glucose levels.
- [“Comparing different supervised machine learning algorithms for disease prediction”](#)
 - The paper reviews several supervised machine learning algorithms used in disease prediction. It highlights the effectiveness of these algorithms in learning from labelled data and making predictions on new, unseen data. The study analysed 48 articles from databases like Scopus and PubMed, which explored a variety of supervised learning techniques. Among the algorithms compared—Support Vector Machine (SVM), Naïve Bayes (NB), Random Forest (RF), Decision Trees (DT), K-Nearest Neighbour (KNN), Logistic Regression (LR), and Artificial Neural Networks (ANN)—SVM was found to be the most frequently used, appearing in 29 studies, followed by NB in 23 studies. The results showed that Random Forest was the most accurate algorithm in 53% of the studies, while SVM achieved high accuracy in 41% of the studies it was applied to. SVM demonstrated its strength in predicting conditions like heart disease, diabetes, and Parkinson’s disease, while ANN was particularly effective for breast cancer prediction. The study emphasizes the importance of selecting the right algorithm based on the specific disease being predicted and helps guide researchers in choosing the most effective method for their work.

Data Sources & Description

We have chosen the diabetes_012_health_indicators_BRFSS2015 dataset from [Diabetes Health Indicators Dataset \(Kaggle\)](#) for our analysis:

- Target Variable: Diabetes_012:
 - Class 0: No diabetes or diabetes only during pregnancy.
 - Class 1: Prediabetes.
 - Class 2: Diabetes.

- The dataset contains 21 feature variables that are collected through a survey, which includes health indicators:
- With 253,680 survey responses, the dataset provides a sufficient amount of data, which is crucial for training machine learning models. The large sample size also improves the model's ability to generalize to real-world scenarios.

Feature Variable	Description	Values
Diabetes_012	Diabetes status	0, 1, 2
HighBP	High blood pressure	0, 1
HighChol	High cholesterol	0, 1
CholCheck	Cholesterol checked	0, 1
BMI	Body Mass Index	Continuous
Smoker	Smoking status	0, 1
Stroke	Stroke history	0, 1
HeartDiseaseorAttack	Heart disease or attack history	0, 1
PhysActivity	Physical activity	0, 1
Fruits	Fruit consumption	0, 1
Veggies	Vegetable consumption	0, 1
HvyAlcoholConsump	Heavy alcohol consumption	0, 1
AnyHealthcare	Healthcare coverage	0, 1
NoDocbcCost	No doctor visit due to cost	0, 1
GenHlth	General health	Categorical (1 – 5 values)
MentHlth	Mental health	Continuous
PhysHlth	Physical health	Continuous
DiffWalk	Difficulty walking	0, 1
Sex	Sex	0, 1
Age	Age	Categorical (13 values)
Education	Education level	Categorical (1-6 values)
Income	Income level	Categorical (1 -8 values)

- This dataset is based on the CDC's Behavioral Risk Factor Surveillance System (BRFSS), which is a reliable source for public health data. It reflects real-world conditions and provides insights into the risk factors associated with diabetes in the population.
- There is an imbalance between the classes, with more instances of Class 0 (no diabetes) compared to Class 1 (prediabetes) and Class 2 (diabetes). This could lead to challenges when building models and may require techniques like resampling or using class weights.
- Reference Dataset: [pima-indians-diabetes-database](#)

Data Processing & Pipeline

- Data Cleaning: Handle missing values, remove inconsistencies.
- Feature Engineering: Normalize numerical variables, encode categorical variables.
- Outlier Detection: Identify and handle extreme values.
- Class Imbalance Handling: Address class imbalance using techniques like resampling (oversampling the minority class or undersampling the majority class) or applying class weights in the machine learning models.

Model Selection

- Feature Selection: Use correlation analysis and mutual information.
- Classification Models: Logistic Regression, Decision Trees, Random Forests, Neural Networks.
- Baseline Model: Logistic Regression for interpretability.

Software & Tools

- R Libraries: tidyverse (data manipulation), caret (machine learning), randomForest, glmnet (logistic regression), ggplot2 (visualization).
- Visualization: ggplot2, corrplot.
- Model Optimization: caret::trainControl, tuneGrid for hyperparameter tuning